DIPLOMA PROGRAMME PRACTICE · SL · SET B

Biology / Paper 2

Worked mark scheme · 50 marks · 90 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each. Answer all of Section A and one of two Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 1 [12]

Inferring grazing effects in a rocky pool

(a) Calculate the difference between treatments in change from start to week 4, and give its units. [3]

1 Exclusion change=42−18=24 g m^{-2}.

1 Open-sided change=24−19=5 g m^{-2}.

1 Difference in changes=19 g m^{-2}.

(b) Explain why the open-sided treatment is more informative than simply comparing a caged plot with a bare rock. [2]

1 The open-sided cage controls partly for shading or physical disturbance from the cage.

1 Grazers still enter, so the comparison more closely isolates the effect of excluding grazers.

(c) Explain why the pool is a sensible unit for pairing plots and identify the independent replicate count. [3]

1 Plots within one pool share local conditions such as wave exposure or nutrient supply.

1 Comparing within pools controls some between-pool variation.

1 There are ten independent pool pairs, not twenty independent pairs.

(d) State a conclusion supported by the means and one measurement needed to assess consistency among pools. [2]

1 Exclusion is associated with greater biomass accumulation over four weeks.

1 Report differences for each pair or their variability/confidence interval to assess consistency.

(e) Suggest a second mechanism besides grazing that could differ between cage types and a way to test it. [2]

1 Closed sides might change water flow or sedimentation.

1 Measure local flow or sediment deposition with matched instruments or add a design isolating this effect.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 2 [8]

Two plant glucose polymers

(a) Identify S and W and distinguish their glucose building blocks. [3]

1 S is starch.

1 W is cellulose.

1 Starch uses α-glucose and cellulose β-glucose.

(b) Connect the arrangement of cellulose chains to wall strength. [3]

1 Alternating glucose orientation produces long straight chains.

1 Hydrogen bonds form between adjacent chains.

1 Bundles or microfibrils resist tensile stress.

(c) Explain why storing glucose as starch helps maintain cellular water balance. [2]

1 A large insoluble polymer contributes fewer osmotically active dissolved particles than many free glucose molecules.

1 This limits osmotic water entry compared with storing the same glucose units as free solute.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 3 [7]

Interpreting vaccination outcomes

(a) Calculate risk in each group and the relative risk reduction. [3]

1 Vaccinated risk=24/800=3.0% and comparison risk=64/800=8.0%.

1 Relative risk=0.03/0.08=0.375.

1 Relative risk reduction=(1−0.375)×100=62.5%.

(b) Distinguish that reduction from the absolute change in risk. State why random assignment matters. [2]

1 Absolute reduction=8.0%−3.0%=5.0 percentage points.

1 Random assignment helps balance prior exposure, age and other confounders between groups.

(c) Explain how vaccination can reduce disease despite some vaccinated participants still developing it. [2]

1 Antigen exposure can generate specific immune memory and a faster response on infection.

1 Individuals vary in response and protection can be incomplete, so reduced risk allows breakthrough cases.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 4 [7]

Reading a sequence of cell images

(a) Describe two changes visible between the first and last images. [2]

1 The cell outline becomes constricted near its middle.

1 Two approximately rounded cell bodies become distinguishable where there was one.

(b) Identify the division process directly supported by the shape change and distinguish it from division of a nucleus. [2]

1 Cytokinesis partitions the cytoplasm, here through cleavage.

1 Mitosis separates duplicated chromosomes into daughter nuclei.

(c) A student claims the images prove the daughter cells have identical DNA sequences. Evaluate the claim and propose evidence. [3]

1 Chromosomes or DNA sequences cannot be compared at this image resolution.

1 Normal mitosis predicts similar nuclear genomes but mutation or segregation errors cannot be excluded from shapes.

1 Compare DNA sequences of the two cells to test sequence identity, or use chromosome analysis specifically to test chromosome complement.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 5 [16]

Supplying an exercising muscle

(a) Outline the main mechanical changes during inspiration in a human at rest. [3]

1 The diaphragm contracts and flattens.

1 External intercostal contraction raises the ribs, increasing thoracic volume.

1 Pressure in the lungs falls below atmospheric pressure, causing inward airflow.

(b) Explain how the alveolar exchange surface allows rapid transfer of oxygen into blood. [5]

1 Many alveoli provide a large surface area.

1 Thin alveolar walls and thin capillary walls create a short diffusion distance.

1 A moist surface permits oxygen to dissolve.

1 Ventilation replenishes alveolar oxygen.

1 Blood flow removes oxygenated blood and maintains a partial-pressure gradient for diffusion.

Paper 2 · Question 5 (continued)

(c) Explain how increased oxygen delivery supports sustained movement, linking transport to energy transfer. [7]

1 Oxygen binds reversibly to haemoglobin in red cells.

1 The left heart pumps oxygenated blood into systemic arteries.

1 Blood reaches capillaries near muscle fibres.

1 Respiring fibres maintain a lower oxygen concentration or partial pressure than incoming blood.

1 Oxygen dissociates and diffuses into the tissues.

1 Aerobic respiration transfers chemical energy from organic substrates into ATP with a high yield compared with anaerobic release.

1 ATP supplies energy for contraction; continuing delivery and substrate supply support repeated cycles.

(Quality) Quality of the complete extended response [1]

1 Award one mark when the complete response communicates the relevant biological reasoning clearly and concisely, with logical connections and little repetition. Assess this across the parts, independently of the content score.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 6 [16]

Variation, selection and conservation

(a) Outline sources of genetic variation in sexually reproducing organisms. [3]

1 Mutation creates new alleles.

1 Meiosis produces new allele combinations by independent assortment or crossing over.

1 Random union of gametes creates additional combinations.

(b) Explain how directional selection could change the trait distribution over many generations. [5]

1 Individuals initially differ in the trait.

1 Some variants tolerate the hot surface better.

1 They survive or reproduce more successfully under those conditions.

1 Their heritable variants are transmitted disproportionately to offspring.

1 The advantageous variants increase in frequency across generations if the selective conditions persist.

(c) Discuss choices that would help the second population retain diversity and allow the conservation team to assess its success. [7]

1 Choose founders across the source population rather than a single family.

1 Include enough unrelated breeders to represent multiple alleles.

1 A small founder sample can lose rare alleles by chance.

1 Small subsequent populations can lose further variation by drift or inbreeding.

1 Provide suitable habitat and resources for successful reproduction.

1 Monitor survival and recruitment over several generations rather than counting released adults alone.

1 Compare genetic diversity or allele frequencies over time and adapt management if diversity declines.

Paper 2 · Question 6 (continued)

(Quality) Quality of the complete extended response [1]

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